Audits, Testing and Fact-Checking

The checks that stand between a change and a node, the tests that exercise the servers and the audio, the audits a node runs on itself, how the summaries in this series were checked against the code, and a plan for finding a problem.

CI CHECKS | TEST SUITES | NODE AUDITS | FIND A PROBLEM

SUMMARY · READ-ONLY REVIEW

Sources: ci/, .github/workflows/, channels/, the tool files, and the checks run for this summary on branch ccr-9d4fe19f-ssud83.

Five places a problem can be caught

LAYER	WHEN	WHAT IT IS
1. Local checks	Before a push	`ci/check_repo.py`, `py_compile` and `pyflakes` on changed Python, `node --check` on page script, `bash -n` on shell, and a test of the changed behaviour (CLAUDE.md)
2. The `checks` workflow	Every pull request	The same repo checks, plus the test suites in Part 3
3. The release gate	Daily and on each push to main	The `promote-stable` workflow decides what reaches the Stable channel (Part 4)
4. Run-time guards	On a node, at the moment of a change	Confirmation, gates, locks and rollback in the tools (Part 5)
5. Node audits and health readings	On a node, any time	The SysMon audit, health verdicts, the boot report, the Suite Log (Part 5)

The first three layers look at the code. The last two look at a running node. A problem found on a node should become a test or a check, so the next change cannot bring it back.

What check_repo.py enforces

The script compares the changed files with the base branch and fails the pull request on any of these. It prints each problem, or "OK" with the number of files checked.

GROUP	RULE
Code	Every changed Python file compiles and passes `pyflakes`. Page script inside a Python file is extracted and parsed with `node --check` (module scripts as modules). Shell files pass `bash -n`.
Versions	One file per tool. The file name and the version inside agree. Three parts for a release, four for a cosmetic build. The pi02w suffix is present or absent as the tool requires. A new version is newer than the previous file. A cosmetic build follows its release and may keep its file name only if the code did not change, only text.
Records	The README lists the file. The tool's changelog has an entry and its "Current file" line is right.
Shared code	Each function in `shared/confio.py` is identical in every tool its `USED_BY` table lists. The shared look (font and colour lines, welcome and licence text) is present in the eight web tools. The launcher text is identical wherever it is written.
Channels	`channels/stable.json` may not be edited by a pull request. The other channel files must have the right shape.
Android	A change under `android/` raises the app version and adds a changelog entry.

The checks cannot judge whether a change is correct. They keep the files, versions and copies consistent, so a node can mix versions safely.

What the tests exercise

SUITE	WHAT IT PROVES	IN THE WORKFLOW	RESULT WHEN RUN FOR THIS SUMMARY
`ci/test_ci.py`	241 tests: the repo checker, promotion rules, atomic writes, Confgetter and Confputter, Node Talker cards and behaviour, receive-health readings, the link and push-to-talk logs, listen events, and the Suite Log	Yes	241 passed
`test_talker_tx_baseline.py`	Transmit audio reaches the USRP port at the expected level, compared with a stored baseline (`ci/fixtures`)	Yes	2 passed
`test_noderecorder_raw.py`	Node Recorder's raw audio path	Yes	3 passed
`test_talker_baseline.py`	Receive audio through the cleanup stage: tone levels, onset latency and signal cycles against a stored baseline	Yes	3 passed (47 s)
`test_servers.py`	Eight servers (Dashboard, pi02w Dashboard, SysMon, pi02w SysMon, Node Talker, amp Node Talker, Node Recorder, Pibrary) start with stubbed hardware; instmon, wifimon and 44helper are not among them. The workflow step is still labelled "six servers". Every route found in the source refuses without a login. Health answers; unknown paths, bad logins and a forged cookie are refused; six wrong logins cause a lockout.	Yes (as root)	72 passed
`test_pages.py`	Six pages (both Dashboards, both SysMons, Node Talker, Node Recorder, before and after login) load in headless Chromium with no script or console errors, has the right title and tab set, and each clicked tab shows its panel	Yes	Not run here (needs a root password set for the tests)
`test_authed.py`	With a login, on the eight test servers except Pibrary: every route, and "badge truth", meaning what a status badge shows matches the service state that the test sets	Yes	Not run here (same reason)
`test_talker_stream.py`	34 tests of Node Talker: login and events (including the X-Requested-With refusal), WebSocket handshake and origin checks, idle close, listen streams, stalled listeners, silence, and a soak test (skipped unless asked for)	**No.** The baseline suites import it only for its helpers, so its own tests never run in CI	34 run: 33 passed, 1 skipped (soak), 3.6 minutes

`test_security.py` **(new)**	For each of the eleven servers: the response headers, the request-shape guard on login, the loopback-proxy rule, the Dashboard's local-API key, and (logged in) the guard on a protected POST, wifimon's proxy refusal and the 44helper terminal's proxy refusal	Yes	25 passed without a login; the logged-in checks run in CI
`test_install.py` **(new)**	Every pi02w and common tool installs and uninstalls cleanly: unit file, launcher, command link, and the launcher removed with the last unit	Yes (opt-in, as root)	Passed for all eight tools
Android job	Unit tests, a release build and lint of the app	Separate workflow	Not run here

Results come from runs on this branch under Python 3.13 (CI uses 3.11). The Chromium suites set the root password, so they were left to CI. `check_repo.py --base origin/main` passed: 108 changed files.

From main to Stable

- `main` is the Testing channel (opt-in on nodes). Stable is the default, and only the `promote-stable` workflow writes it.
- The rules are in `channels/rules.json`: a release becomes stable once 4 newer releases of that tool are on main and it has been there 3 days, or once it has been the newest release for 14 days. Cosmetic builds do not count. A floor sets the lowest version a tool may be promoted at.
- `channels/overrides.json` (changed only by pull request) can hold a version back or send a hotfix to Stable at once. The file today holds back versions of wifimon, 44helper, Node Talker and instmon, and hotfixes versions of instmon and pibrary.
- The workflow runs daily and on every push to main, and retries its push up to three times.
- On a node, instmon checks each download against its git blob id, so a changed file is refused (instmon part 1 of this series).

The delay is the audit: a bad release has days on Testing to be noticed, and a hold in `overrides.json` stops it reaching Stable.

Guards, audits and health readings

5.1 Guards that act before a change

WHERE	GUARD
instmon	Confirmation on installs, scripts and updates; a syntax check and a self-install check before an install; no action during a Full Update; actions locked while Node Recorder is recording; one install per tool; library locks; a 60-second stay-up watch after an install; Full Update rollback, and a 90-second timer that restores instmon itself.
44helper	The command a step runs must equal the command in its table; a masked-service gate; a broken-package gate (`dpkg --audit`); field rules; the terminal's switch, address, origin, password and ticket checks; a 15-minute auto-undo on the tunnel tests.
Node Controller	One link lock for all connects; the bridge reaper and watchdog; a boot disconnect that waits for Asterisk.
Node Talker	A key-up refused when the node is not ready, not linked or busy; a time-out and a dead-man stop; a load guard that stops tests when the Pi is stressed.
wifimon	Watchdogs for Wi-Fi, voltage and the tunnel; a test of a new network with undo; a control gate on changes.

5.2 Audits and readings

READING	WHERE	WHAT IT TELLS YOU
Security audit, 14 checks	Full-build SysMon only	Port exposure, default credentials, SSH, Fail2ban, Monit, service user, file permissions, and more. **The pi02w build has no audit.**
Self-check and readiness	44helper	The tunnel interface, `wg0.conf` permissions, firewall exposure, and whether each setup step is really done
Boot report	44helper (Boot tab, `--boot-report`)	Uptime, clock sync, power, journal warnings, restarted or never-started units; saved 6 and 25 minutes after each boot
Receive-health verdict	Node Talker	Which layer lost audio: the network outside the LAN, the Pi, the Pi's Wi-Fi, the link to the listener, or the phone
Audio check, tone test, kerchunk test	Node Talker	Measured levels, drop-outs, delay and bandwidth; rules that suggest a setting change
Suite Log and coverage	instmon	A redacted record of journals, files, Wi-Fi and 10-minute snapshots, and a coverage table showing which expected sources produced rows
Quick tests and conflicts	SysMon	For each pinned service: installed, active, port listening, config present; and duplicate ports
Shutdown reports	wifimon	The newest 10 reports from watchdog shutdowns

How the summaries were checked against the code

Each of the 26 summaries was read against the source. The methods below found real errors, and each can be repeated.

METHOD	WHAT IT CATCHES	EXAMPLE FROM THIS SERIES
Search for the constant or string	A wrong number, name or path	Lockout starts at the fifth failure, not the first; each tool has its own cookie name
Load the module and count its tables	Counts that were estimated by eye	The 44helper step tables: ASL3 has 16 install steps, not 15; Ampersand 7 and 6, not 12
Compare function trees (`ast`)	A shared function that has drifted	All 52 copies of the 13 shared functions match
Hash extracted text	A copied block that differs	The launcher is identical in all 11 files
Parse the markup with an HTML parser	Counts distorted by script text	Control and ARIA counts per page, corrected
Follow a call to its end	A claim about a feature that the code does not support	Phone tones are sent by the Dashboard, not by Node Talker; the kerchunk log is kept in the Suite Log file
Read the gate, not the label	A name that sounds like a rule	Pibrary's "access" and SysMon's "lan" flag only label a client; they refuse nothing
Search the whole repo for what is absent	A missing protection	No sandbox directive in any unit; five tools serve plain HTTP; the pi02w SysMon has no audit
Render and look	Layout faults	A wrong footer label, and pages holding one stray line

6.1 Kinds of error found

- Numbers and names that were right once and drifted (counts, limits, file names).
- A claim taken from a manual and not checked in code, or the reverse.
- A feature described from its label, when its code did something narrower.
- Something real that the first reading missed (connection caps, the shared login file, the custom command mode).

Summaries mark what was not checked. Those lists are the next place to look.

A plan, in order

7.1 In the repo

1. Run `python3 ci/check_repo.py --base origin/main`. Fix every line it prints.
2. Run `python3 ci/test_ci.py`, then the baseline and server tests. A failure in a baseline means the audio changed.
3. For a page change, run the page and authenticated tests on a runner that has Chromium, or open the page headless and watch the console.
4. Run `pyflakes` and `node --check` on what you touched, and exercise the changed behaviour.
5. Search for the name you changed in every tool: shared functions and the launcher live in many files, and the full build repeats the pi02w code.

7.2 On a node

1. Look at the Boot tab or run `44helper.py --boot-report --json`: clock, power, restarted units.
2. Look at SysMon's quick tests and port conflicts, and (full build) the audit.
3. For audio, read Node Talker's health verdict first. It names the layer that lost audio.
4. Start an instmon Suite Log collection, reproduce the fault, and read the coverage table for silent sources.
5. Check the tool's own log and the service journal; look for lockouts, refused requests and rollbacks.

7.3 Symptom to place

SYMPTOM	LOOK AT
A tool will not update or install	instmon log; the channel and its stable list; a hold in `overrides.json`; a lock or an install already running
A link will not make or drop	The Dashboard link lock and its "Busy" message; the bridge reaper; Node Talker's guard (it acts only when it owns the node)
Audio drops or is late	Node Talker's verdict row by row, then the load guard and the Pi's power flags
A key-up is refused	The reason code: not-ready, not-linked, busy, in-use (TX path)
The node rebooted or shut down	wifimon: the network-loss action, the voltage watchdog and its shutdown reports
The tunnel is not working	The tunnel guard's state file, the handshake, the boot order units, and any half or exceptions test still armed
A page will not load or log in	The lockout (15 minutes or a doubling wait), the shared Dashboard and SysMon session, a proxy header refusal, the connection cap
A setting has no effect	Node Talker's level switch; whether the tool was restarted (launcher flags are read at start)

What is missing, and what to add

Each gap below was checked in the repository. The status column says what was applied in this round; the rules that keep them applied are in the Testing and audit policy in CLAUDE.md.

#	GAP (EVIDENCE)	FIX	STATUS
1	The Node Talker stream tests (34) never run in CI. The workflow has no step for `test_talker_stream.py`.	Add a step to `checks.yml`: `python3 ci/test_talker_stream.py` (about 4 minutes; the soak stays off).	Applied
2	instmon, wifimon and 44helper have no server tests: no check that their routes refuse without a login, that a wrong login is refused, or that they lock out.	Add the three to `SERVERS` in `ci/test_servers.py` with temporary config folders (instmon reads `INSTMON_CONFIG_DIR` and `INSTMON_LIBRARY_DIR`; wifimon and 44helper use fixed folders, so they need a folder override or the runner's own paths). Then add them to the authenticated and page tests, with Pibrary.	Applied to the server, security and install tests (eleven servers). Page and logged-in tests for the three are still to add
3	No test sends a request that should be refused for its shape: a POST with no `X-Requested-With` (or, in the Dashboard and SysMon, no JSON type), a request with a proxy header to wifimon or the 44helper terminal, a local-API call without the key or from a non-loopback address, or a WebSocket with a foreign Origin (only Node Talker's is tested).	A new `ci/test_security.py` driven by a table of expected answers per tool: 403 or 415 for the missing header or type; 403 for a proxy header; 403 for the local API without the key. Run it against the same stub servers.	Applied (`test_security.py`). Found: instmon and Pibrary take a login without the header; recorded as a known gap
4	Response headers are not tested. The security summary lists which tool sends which header, but nothing keeps it true.	In `ci/test_security.py`, fetch each page and compare `X-Frame-Options`, `nosniff`, `Referrer-Policy` and `Content-Security-Policy` with a table, so a missing header fails CI.	Applied
5	The workflow runs on pull requests only, not on pushes to main, so the merged result is never tested.	Add `push: branches: [main]` to `checks.yml`.	Applied
6	CI uses Python 3.11 only. Debian 13 (Trixie) nodes run 3.13.	Run the Python steps on a matrix of 3.11 and 3.13. (The suites passed on 3.13 when run for this summary.)	Applied (matrix 3.11 and 3.13)
7	44helper's exact-command rule, its field rules and its custom mode have no tests. A step field with no rule makes the step unrunnable, and nothing finds that.	Unit tests in `ci/test_ci.py`: for every step table, every field has a rule in `_STEP_FIELD_RULES`; `_step_command` builds the shown command; a changed command is refused; a bad field value is refused.	Applied (8 tests)

8	The comment-stripped rule is not enforced. Today no tool file has a Python comment or docstring (checked with the tokenizer), but `check_repo.py` would not notice one.	In `check_python`, fail on any comment token after the shebang and coding lines, and on any docstring; for shell, allow only the shebang and title line.	Applied (Python and shell)
9	The launcher and `-OO` are never exercised. The test servers start each tool directly.	A test that writes the launcher from a tool file, runs a small target through it, and checks the re-run keeps the process ID, applies `-OO` and the arena cap, removes the cap from child processes, honours each `launch_no_*` flag, and loads a file whose name starts with a digit. Also start one server-test run through the launcher.	Applied (tests, and servers start through the launcher in CI)
10	`--install` and `--uninstall` are untested: unit files, links, the launcher, firewall ports, and the rule that the launcher is removed only when no unit uses it.	A test that runs each tool's install and uninstall in a temporary root with `systemctl`, `firewall-cmd` and `ufw` replaced by stubs on `PATH`, and checks what was written and removed.	Applied (`test_install.py`)
11	Promotion to Stable does not check that the commit passed CI. `ci/promote.py` reads only the history of main.	Make `promote.yml` skip a release whose commit has no successful `checks` run, or rely on branch protection and say so in the README.	Applied (promotion waits for green checks)
12	Shell scripts get only `bash -n`.	Run `shellcheck -S error` on changed `.sh` files in `check_repo.py`.	Applied (CI requires shellcheck)
13	Only changed files are checked. An unchanged tool is not re-checked when Python or Node changes.	A weekly scheduled run that checks every tool file, not only the changed ones.	Applied (weekly `--all` run)
14	Disk-image backup, restore, clone and format are not tested. instmon accepts only USB or removable drives, so a loop device cannot stand in.	Add a test-only switch (as Pibrary has `PIBRARY_TEST_ALLOW_LOOP`) and test the jobs on loop devices made from image files.	Not applied: needs an instmon release with a test switch
15	Unused code is not found. Two page-script functions in 44helper are never called.	A warning in `check_repo.py` for page-script functions defined and never named, with an allow-list for names built at run time.	Applied as a note; removal goes in 44helper's next release
16	The summaries drift from the code. Each names its source file in the footer, and nothing warns when that file is replaced.	A note in `check_repo.py`: when a tool gets a new file name, list the files in `docs/src` that still name the old one.	Applied as a note
17	No test runs on a real node. Asterisk, DVSwitch, WireGuard, firewalld and the radio path are stubbed.	A short release checklist run on a Pi Zero 2 W before a hold is lifted: link and unlink each mode, a tone test, a tunnel restart, a reboot, and the boot report.	Added to the policy
18	The workflow step says "six servers". There are eight.	Correct the label in `checks.yml`.	Applied

8.1 Product gaps the tests cannot fix

- The pi02w SysMon has no security audit. Only the full build does.

- Five tools serve plain HTTP, and every service runs as root without sandboxing (see the security summary).
- Memory and timing on a Pi Zero 2 W (the launcher savers, the light-theme filter) were not measured.

8.2 Order used

First the workflow-only changes (1, 5, 6, 18); then the server and security tests (2, 3, 4), which cover the largest untested surface; then the small additions (7, 8); then 9 to 12 and the rest.